

Bribe to Cooperate

Zhiyong Yao*

School of Management, Fudan University, Shanghai, China

This article discusses how to sustain R&D cooperation between two firms with different technological capabilities. We show that a long-term relationship combining with transfer can help to maintain mutual cooperation when the discount factor is not sufficiently large. Furthermore, the less capable firm is more likely to deviate because its payoff from the deviation is higher than that of the more capable one. Under certain conditions, the more capable firm can ‘bribe’ the less capable firm to cooperate, and both firms are better off. But the less capable firm receives more. Copyright © 2011 John Wiley & Sons, Ltd.

1. INTRODUCTION

The Matthew effect has long been recognized in scientific research (Merton, 1968) as well as in other areas (Merton, 1988).¹ In the context of R&D cooperation, previous studies also identify the Matthew effect, that is, the more capable or bigger firm obtains a disproportionately larger share from the value created by the successful joint R&D. For example, Sinha and Cusumano (1991) find that big firms have greater incentives to cooperate than small firms, and firms prefer as small a partner as possible to limit the sharing of results. Veugelers and Kesteloot (1996) show that the advantage firm has incentive to join an alliance with a small firm only if its advantage is reflected in a larger share of the benefits created by the alliance.² Fershtman and Kamien (1992) and Perez-Castrillo and Sandonis (1996) also obtain results demonstrating the Matthew effect. However, in the similar R&D cooperation environment, this article presents the opposite effect: the anti-Matthew effect, that is, the less capable firm receives more under certain conditions.

Using a game theoretical framework, this article explores ways to sustain R&D cooperation between asymmetric partners in order to reap the synergy while avoiding moral hazard. In our model, two firms, different in technology capabilities, form a research joint venture (RJV). They first take unobservable actions, either ‘cooperating (high efforts)’ or ‘deviating (low efforts)’, for the RJV, and then the R&D outcome is commonly observed. Because the R&D process is stochastic, the outcome serves as a noisy signal of the actions taken. We find that because the less capable firm is more likely to deviate, the more capable firm can ‘bribe’ it to cooperate by transfer, and both firms will be better off. But the more capable firm receives less. The reason for this result is as follows. In our team work environment, both firms’ participations are essential for the success of the R&D. But the payoff from deviation for the less capable firm is higher than that of the more capable one. Transfer can induce the less capable firm to cooperate by increasing the opportunity cost of its deviation.

We begin our analysis by examining a one-shot game with equal sharing rule. We distinguish the unilateral deviation case and the bilateral deviation case. Because the bilateral deviation case is a standard prisoners’ dilemma, the repeated interactions can help the two firms achieve mutual cooperation. Because the less

*Correspondence to: School of Management, Fudan University, Room 236, Siyuan Building, 670 Guoshun Road, Shanghai, China.
E-mail: yzy@fudan.edu.cn

capable firm is more likely to deviate, it requests a larger threshold discount factor to sustain the cooperation. The most interesting case is when the common discount factor is below the less capable's minimum threshold but above that of the more capable. Under certain conditions, the more capable firm can still attract the less capable firm to cooperate by transfer. The transfer will increase the less capable firm's continuation payoffs and thus decrease its threshold discount factor for cooperation.

In the unilateral deviation case, the more capable firm cooperates, whereas the less capable does not. When the R&D outcome is verifiable, the formal incentive contract can be used to induce the less capable firm to cooperate. The contract specified a transfer from the more capable firm to the less capable one once the R&D is successful. But if the R&D outcome is nonverifiable, the self-enforcing relational contract is an alternative to induce the less capable firm to cooperate. The two firms engage in a long-term relationship, where every period, the more capable firm will make a transfer to the less capable as long as the R&D is successful.

We then endogenize the sharing rule. And we confirm the previous result that in order to maintain mutual cooperation, the less capable firm receives a larger share than the more capable one when the common discount factor is not sufficiently large.

As mentioned previously, papers on R&D cooperation between asymmetric partners point out the Matthew effect (Fershtman and Kamien, 1992; Perez-Castrillo and Sandonis, 1996; Sinha & Cusumano, 1991; Veugelers and Kesteloot, 1996), whereas our paper demonstrates the anti-Matthew effect. The key difference is as follows. All of the previous models assume that the firms can deviate from the RJV by performing independent R&D. The more capable firm is more capable in conducting its own R&D. Thus, it obtains a larger share from the RJV. However, our paper assumes that firms cannot do independent R&D and the more capable firm is more capable in doing joint R&D. In this environment, the less capable has higher incentive to shirk, but its cooperation is essential for the success of the joint R&D. Hence, in order to induce the less capable to cooperate, it obtains more.

Many articles on RJVs assume the spillover effect (e.g., D'Aspremont and Jacquemin, 1988; Kamien *et al.*, 1992; among others), but our model assumes complementarity between cooperative partners.³ Furthermore, the complementarity in our model is so strong that the firm cannot perform the R&D

independently or the cost will be extremely high. There are many such cases in real life. For example, Roijakkers *et al.* (2005) point out that R&D cooperation between large pharmaceutical companies and small biotechnology firms are highly complementary: without the small biotechnology firms, the large pharmaceutical companies do not have the specific discoveries, whereas without the large pharmaceutical companies, the small biotechnology firms cannot develop their discoveries into products. A lot of R&D collaborations among vertically related firms are highly complementary (Banerjee and Lin, 2001).⁴ Therefore, the R&D cooperation in our model is more like the team production in the sense of Alchian and Demsetz (1972).

Our paper is also related to the repeated game literature, especially Harrington (1989) and Fong and Surti (2010). Harrington (1989) studies collusive behavior among firms with different discount factors. He finds that collusion is possible if the *average* discount factor is sufficiently high. Those firms with relative high discount factors have 'excess colluding ability', and they can compensate firms with relatively low discount factors so as to induce them not to deviate from collusion. Hence, the firms with relatively low discount factor receive higher market shares. But the firms with relatively high discount factors still earn higher total discounted payoffs because they are more patient. In our model, the firms have the same discount factors. But the more capable firm has the 'extra cooperating power' because it can afford to cooperate at lower discount factor than the less capable one. When the common discount factor is sufficiently high but below the minimum threshold discount factor of the less capable firm, the more capable firm can induce the less capable firm to cooperate by transfer.

Both Fong and Surti (2010) and our paper combine long-term relationship with transfer to study the prisoners' dilemma game. But as Harrington (1989), Fong and Surti (2010) analyze the game between two players with different discount factors. They distinguish the full cooperation and the partial cooperation cases. In the partial cooperation case, Pareto dominant equilibrium payoffs can be implemented, and the impatient player makes repeated payments to the patient one except the first period. In the full cooperation case, the payments can go either direction depending on the different combinations of the discount factors. Therefore, it is not clear which player receives more in their paper. The main difference

between their model and ours is that their game assumes perfect monitoring, whereas ours assumes imperfect monitoring. There are several papers discussing R&D cooperation between rivals by repeated game models, for example, Kesteloot and Veugelers (1995) and Eaton and Eswaran (1997). But both of them focus on the symmetric firms with perfect monitoring, which are different from our asymmetric firms with imperfect monitoring scenario.

In our model, transfer combines with the long-term relationship to facilitate cooperation between asymmetric firms. In cartel literature, side payments are often used to facilitate collusions (e.g., McAfee and McMillan, 1992). Also in cartel practice, when direct transfer is not allowed, side payments can also be implemented by a firm buying output from another firm (see Harrington and Skrzypacz, 2007). In the scenario of R&D cooperation between asymmetric partners, the side payments or transfer can also take all kinds of forms, such as extra contributions or funds to the RJV, asymmetric distribution of benefits or costs, and so on. Our research calls for empirical works on this regard.

The structure of this article is as follows. Section 2 describes the model and discusses the one-shot game. Section 3 deals with the bilateral deviation case. Section 4 analyzes the unilateral deviation case. Section 5 endogenizes the sharing rule. And finally, Section 6 concludes.

2. MODEL

Two risk-neutral firms, A and B, set up a joint venture in order to engage in R&D cooperation. They simultaneously choose actions C or D for the RJV, where C stands for ‘high efforts (cooperation)’ and D for ‘low efforts (deviation)’. The costs for actions C and D are normalized to c and 0, respectively, for both firms. The outcome for the RJV is binary, either ‘good’ or ‘bad’. If the outcome is ‘good’, an innovation of value 2π is achieved; if it is ‘bad’, the payoff is 0. The outcome of the project stochastically depends on the actions by the firms. If both firms choose C , the probability of ‘good’ outcome is γ . If firm i takes action C , whereas the other firm chooses D , the probability of ‘good’ outcome is η_i , where $i = A, B$. We assume $\eta_A > \eta_B > 0$. In other words, firm A is more capable in R&D than firm B. If both firms choose D , the probability of ‘good’ result is 0. The outcome of the project is public observable by both firms, but their actions are not. We assume further that $\eta_A + \eta_B < \gamma < 1$. In this assumption, $\eta_A + \eta_B < \gamma$ represents the synergy effect by

combining the firms’ complementary R&D efforts or resources. But the synergy effect cannot make the R&D successful for sure, that is, $\gamma < 1$. In addition, we assume that the R&D needs team work. Neither firm can develop the new technology on its own house because the other party’s presence is essential even though it chooses D .

2.1. One-shot Game with Equal Share

Suppose the innovation is successful, the two firms share the value created equally, and each firm gets payoff π . The equal sharing rule is often empirically observed (Veugelers and Kesteloot, 1996), and it is also adopted by other papers (Silipo, 2005). It serves as the benchmark in our model, and we will endogenize the sharing rule later.⁵

First, we examine the one-shot game. The firms’ strategic interaction can be illustrated by the following normal-form game:

		Firm B	
		C	D
Firm A	C	$\gamma\pi - c, \gamma\pi - c$	$\eta_A\pi - c, \eta_A\pi$
	D	$\eta_B\pi, \eta_B\pi - c$	0, 0

We further assume that $\gamma < 2\eta_A$.⁶ Therefore, we can summarize all the possibilities into the following three cases:

Case 1. The cooperation case. When $\frac{c}{\pi} \leq \gamma - \eta_A$, or the cost-benefit ratio $\frac{c}{\pi}$ is sufficiently small, choosing C is the dominant strategy for firm A. Given firm A choosing C , firm B chooses C as well. Therefore, the Nash equilibrium for the stage game is (C, C) .

Case 2. The unilateral deviation case. When $\gamma - \eta_A < \frac{c}{\pi} \leq \eta_A$, firm A has dominant strategy C , and firm B has dominant strategy D . Therefore, the Nash equilibrium for the stage game is (C, D) . In this case, firm B is a free rider. But when $\frac{c}{\pi} < \min\{2(\gamma - \eta_A), \eta_A\}$, the total payoffs when both firms choosing C are greater than those of Nash equilibrium (C, D) .

Case 3. The bilateral deviation case. When $\eta_A < \frac{c}{\pi} < \gamma$, firm B has dominant strategy D . Given firm B choosing D , firm A chooses D . Therefore, the Nash equilibrium for the stage game is (D, D) . This is the prisoners’ dilemma case. In this case, if both firms choose C rather than D , both firms can be better off since $\gamma\pi - c > 0$.

With the equal sharing rule, under all possible parameter values for $\frac{c}{\pi}$, the less capable firm is more likely to deviate than the more capable one. In the unilateral deviation case and the bilateral deviation case, as we have shown, the total payoffs would be higher if both firms had chosen C . And both parties could be potentially better off under mutual cooperation. Hence, in the following two sections, we will explore alternative ways in order to obtain cooperation in both cases.

3. THE BILATERAL DEVIATION CASE

Here, we examine the bilateral deviation case, in which the Nash equilibrium for the one-shot game is (D, D) . Because it is a standard prisoners' dilemma, the two firms can engage in repeated interactions in order to obtain the mutual cooperation solution.⁷

3.1. Repeated Game

The repeated game is an infinite repetition of the one-shot game described in Section 2.1. At the beginning of each period $t=0, 1, 2, \dots$, both firms simultaneously choose actions C or D for the RJV. At the end of each period, they can observe the R&D outcome, $y^t \in \{\text{good}, \text{bad}\}$. The public history at the beginning of period t is $h^t \equiv (y^0, \dots, y^{t-1})$. A public strategy σ_i for firm i is a sequence of functions $\{\sigma_i^t\}_{t=0}^\infty$, where each σ_i^t maps each h^t to the action set $\{C, D\}$. At $t=0$, the null history is mapped to the action set $\{C, D\}$. The public strategy depends only on the public history h^t and not on any private history. Firm i maximizes the normalized discounted sum of payoffs:

$$v_i(\sigma) = (1 - \delta) \sum_{t=0}^{\infty} \delta^t u_i(a_i^t, a_j^t),$$

where $a_i \in \{C, D\}$ and $u_i(a_i^t, a_j^t)$ is firm i 's payoff at period t , $i = A, B$.

The equilibrium concept we use here is perfect public equilibrium (PPE henceforth). A PPE is a profile of public strategies that constitutes a Nash equilibrium in the continuation game at any date t , given any public histories.⁸

We first specify a cooperative strategy in the spirit of Green and Porter (1984): The game starts in a cooperation phase in which both firms choose C . They continue doing so as long as 'good' outcome is observed in the previous period. However, once 'bad' outcome is observed, the game enters into a punishment phase in

which both firms choose D for T periods. After the punishment phase, a new cooperation phase starts again with both firms choosing C . We can obtain the following result:

Proposition 1:

There exists a pair $(\bar{\delta}, T)$ such that for all $\delta \geq \bar{\delta}$, the cooperative strategy constitutes a PPE of the repeated game.

Proof. See Appendix.

When firms engage in long-term R&D cooperation, the expected future payoffs act as the 'carrot and stick' that can discipline firms' behaviors. In the cooperation phase, whereas choosing D increases the current payoff, it also increases the possibility of entering a punishment phase in which firms' payoffs will be lower. If firms are patient enough, choosing D will not be the best response because the expected loss in future payoffs due to mutual punishment exceeds the gain from choosing D . In Proposition 1, we find the threshold discount factor $\bar{\delta}$ such that when δ is larger than or equal to $\bar{\delta}$, both firms will stick to the cooperative strategy specified previously.

Note that because of the stochastic nature of R&D, it will fail even if both firms choose 'C'. Thus, noncooperation happens in the equilibrium path occasionally. However, it does not mean firms' inability to cooperate on R&D. Rather, the use of punishment phase helps to ensure cooperation in the cooperation phase. Empirical research needs to distinguish whether the cooperation is stopped temporarily or terminated forever.⁹

Now we proceed to find the minimum threshold discount factor to support the equilibrium. Note that in our setting, T stands for the severity of punishment. The larger T is the lower $\bar{\delta}$ that is needed to ensure cooperation. Therefore, to determine the minimum threshold discount factor, we let $T = \infty$. This is also corresponding to the grim trigger strategy: The game starts in a cooperation in which both firms choose C for the RJV. They continue doing so as long as 'good' outcome of the RJV is observed in the previous period. Once 'bad' outcome is observed, both firms choose D forever.

Under the grim trigger strategy, the equilibrium payoff vector, denoted by $V_i = \{V_A, V_B\}$, can be written as follows:

$$V_i = (1 - \delta)(\gamma\pi - c) + \delta\gamma V_i. \quad (1)$$

If firm i deviates, its payoff is

$$V_i' = (1 - \delta)\eta_j\pi + \delta\eta_j V_i,$$

where $i = A, B$; $i \neq j$.

For the grim trigger strategy to be an equilibrium, we need $V_i \geq V'_i$, or

$$(1 - \delta) \left[c - (\gamma - \eta_j) \pi \right] \leq \delta (\gamma - \eta_j) V_i. \quad (\text{IC})$$

Combining (IC) with Eq. (1), we have

$$\delta_i^* \equiv \frac{c - (\gamma - \eta_j) \pi}{c \eta_j}.$$

It is easy to show that $\delta_A^* < \delta_B^*$. The reason is because the less capable firm B has a higher payoff from deviation than firm A. Thus, it is more likely to deviate. In order to sustain the cooperation behavior, it requests a higher threshold discount factor.

For all $\delta \geq \delta_B^*$, the grim trigger strategy is a PPE of the repeated game. Meanwhile, when $\eta_A < \frac{c}{\pi} < \frac{\gamma - \eta_A}{1 - \eta_A}$, we have $0 < \delta_B^* < 1$. Therefore, we have the following Corollary:

Corollary 1:

When $\eta_A < \frac{c}{\pi} < \frac{\gamma - \eta_A}{1 - \eta_A}$, there exists a $\delta_B^* \equiv \frac{c - (\gamma - \eta_A) \pi}{c \eta_A}$ such that for all $\delta \geq \delta_B^*$, the grim trigger strategy is a PPE of the repeated game.

When the common discount factor is large enough, that is, larger than or equal to δ_B^* , mutual cooperation could be supported as the equilibrium outcome as we have shown. However, firms could be less patient because of a variety of reasons. For example, firms may have myopic behaviors due to agency problems, or they may have a high cost of capital and thus a high rate of time preference (Harrington, 1989).

What would happen if firms are less patient? Could the mutual cooperation be obtained when the common discount factor is smaller than the threshold, or $\delta < \delta_B^*$? In the symmetric firm case, there is no way to support cooperation outcome under this situation. However, in our asymmetric firm case, it is possible because firm A is willing to cooperate at a lower discount factor than the threshold, as we have shown $\delta_A^* < \delta_B^*$. If the common discount factor is in between of δ_A^* and δ_B^* , or $\delta_A^* < \delta < \delta_B^*$, and the benefit from cooperation is large enough, firm A can afford to ‘bribe’ firm B to cooperate.

3.2. Long-term Relationship with Transfer

Given $\delta_A^* < \delta < \delta_B^*$, in order to achieve the mutual cooperation outcome, we propose a cooperative strategy with transfer. We consider the following grim trigger strategy¹⁰: The game starts in a cooperation in which

both firms choose C for the RJV. If the R&D is successful, firm A makes a transfer τ to firm B. They continue doing so as long as ‘good’ outcome is observed and τ is made in the previous period. However, once ‘bad’ outcome is observed or τ is not made, both firms choose D forever and firm A stops making transfer forever.

Under this strategy profile, firm A makes a transfer τ to firm B once the R&D is successful. By making such a transfer, the continuation payoff for firm A will decrease, whereas the continuation payoff for firm B will increase. Then the initial threshold discount factor for firm A to cooperate, δ_A^* , will increase, and the initial threshold discount factor for firm B to cooperate, δ_B^* , will decrease. Now we proceed to identify the conditions under which the cooperative strategy with transfer τ is a PPE.

The equilibrium payoff vector $V_i^\tau = \{V_A^\tau, V_B^\tau\}$ can be respectively written as

$$V_A^\tau = (1 - \delta)[\gamma(\pi - \tau) - c] + \delta\gamma V_A^\tau, \quad (2)$$

and

$$V_B^\tau = (1 - \delta)[\gamma(\pi + \tau) - c] + \delta\gamma V_B^\tau. \quad (3)$$

Firm A can deviate from the cooperative strategy by two ways. One is it chooses D but makes the transfer τ to firm B if the R&D is successful; the other is choosing C but stopping the transfer τ given the R&D is successful. The payoff for the first kind of deviation is

$$V_A^{\tau'} = (1 - \delta)\eta_B(\pi - \tau) + \delta\eta_B V_A^\tau.$$

The expected payoff for firm A for the second kind of deviation is

$$V_A^{\tau''} = (1 - \delta)(\gamma\pi - c).$$

Firm B can deviate by choosing D , and the payoff is

$$V_B^{\tau'} = (1 - \delta)\eta_A(\pi + \tau) + \delta\eta_A V_B^\tau.$$

For the cooperative strategy with transfer τ to be a PPE, the following incentive constraints should be satisfied: $V_A^\tau \geq V_A^{\tau'}$, $V_A^\tau \geq V_A^{\tau''}$, $V_B^\tau \geq V_B^{\tau'}$. By the first two constraints, we obtain $\tau_1 \leq \pi - \frac{c(1 - \delta\eta_B)}{\gamma - \eta_B}$ and $\tau_1' \leq \delta(\gamma\pi - c)$, respectively. And from $V_B^\tau \geq V_B^{\tau'}$, we have $\tau \geq \frac{(1 - \delta\eta_A)c}{\gamma - \eta_A} - \pi$.

To guarantee that both firm A and firm B will follow the cooperative strategy specified previously, and to maximize V_A^τ , we let

$$\tau^* = \frac{(1 - \delta\eta_A)c}{\gamma - \eta_A} - \pi, \quad (4)$$

and τ^* should also satisfy $0 < \tau^* \leq \min\{\tau_1, \tau_1'\}$, which requires $\delta' \leq \delta < \delta_B^*$, where $\delta' \equiv \frac{(\gamma - \eta_B)[c - \pi(\gamma - \eta_A)] + (\gamma - \eta_A)[c - \pi(\gamma - \eta_B)]}{c[\eta_A(\gamma - \eta_B) + \eta_B(\gamma - \eta_A)]} > \delta_A^*$. The common discount factor δ should be large enough, that is, $\delta \geq \delta'$; otherwise, we cannot find τ^* defined in Equation (4). Furthermore, given $\delta' \leq \delta < \delta_B^*$, the larger the δ is, the smaller the τ^* that is needed because $\frac{d\tau^*}{d\delta} < 0$. Then we have the following Proposition:

Proposition 2:

Given $\delta' \leq \delta < \delta_B^*$, there exists τ^* defined in Eq. (4) such that the grim trigger strategy with transfer τ^* can be supported as a PPE of the repeated game.

Because the bilateral deviation case is a standard prisoners' dilemma, it is well-known that if the two firms are patient enough, they can engage in a long-term relationship to achieve the beneficial cooperation outcome. Because of the asymmetry, the two firms' minimum threshold discount factors to sustain the cooperation are different. Because the less capable firm is more likely to deviate, it requests a larger threshold discount factor. In the case that the common discount factor is below the less capable's minimum threshold but above that of the more capable, under certain conditions specified in Proposition 2, the more capable firm can attract the less capable firm to cooperate by transfer τ^* . The transfer will increase the less capable firm's continuation payoffs and thus decrease its threshold discount factor for cooperation. Both firms are better off, but the less capable firm achieves more than the more capable one.

4. THE UNILATERAL DEVIATION CASE

In the unilateral deviation case, firm A chooses C , whereas firm B chooses D . However, when $\frac{c}{\pi} < \min\{2(\gamma - \eta_A), \eta_A\}$, the total payoffs when both firms choosing C are larger than those of Nash equilibrium (C, D). If firm A can compensate firm B enough, both firms can be potentially better off by moving from Nash equilibrium (C, D) to new equilibrium (C, C). To achieve this goal, firm A may give firm B a transfer after dividing the benefit equally. Or equivalently, firm

A may allow firm B to take a larger share of the innovation benefit.

4.1. Formal Contract

Given the R&D outcome is verifiable, the two firms can sign a binding contract. Essentially, the designed contract should satisfy both firms' participation constraints and avoid moral hazard at the same time. Because firm A cannot observe firm B's choice directly, the contract term can only depend on the R&D outcome: If the R&D is a success, after the two parties divide the innovation benefit equally, firm A makes a transfer ω to firm B; if it is a failure, no such transfer will be made.

Formally, firm A will design a contract to solve the following program:

$$\max_{\{\omega\}} \gamma(\pi - \omega) - c$$

$$\gamma(\pi - \omega) - c \geq \eta_A\pi - c, \quad (\text{PC}_A)$$

$$\gamma(\pi + \omega) - c \geq \eta_A\pi, \quad (\text{PC}_B)$$

$$\gamma(\pi + \omega) - c \geq \eta_A(\pi + \omega). \quad (\text{IC}_B)$$

Note that PC_A (PC_B) is the participation constraint for firm A (B), which requires that the contract should offer firm A (B) payoff no less than without it. IC_B is the incentive compatibility constraint for firm B, which ensures that with the transfer, firm B has incentive to choose C rather than D .

It is obvious that once IC_B is satisfied, PC_B is met automatically. When IC_B is binding, we can obtain $\omega^* = \frac{c}{\gamma - \eta_A} - \pi$. It is easy to check when $\omega^* = \frac{c}{\gamma - \eta_A} - \pi$, PC_A is also satisfied. Therefore, under this contract, firm B's expected payoff is $\frac{c\eta_A}{\gamma - \eta_A}$, whereas firm A expects to get $2\pi\gamma - c - \frac{\gamma c}{\gamma - \eta_A}$.

In other words, in order to induce firm B to cooperate, the equal sharing rule should be replaced by the unequal sharing rule, in which firm B takes a larger share once the R&D is successful. Rather than dividing the innovation value equally, firm B takes the proportion $\frac{c}{2\pi(\gamma - \eta_A)}$ of the total payoffs, whereas firm A takes the left, which is $1 - \frac{c}{2\pi(\gamma - \eta_A)}$.

Defining the share rule $v^* \equiv \left(\frac{c}{2\pi(\gamma - \eta_A)}, 1 - \frac{c}{2\pi(\gamma - \eta_A)}\right)$, then we have the following lemma:

Lemma 1:

When $\gamma - \eta_A < \frac{c}{\pi} \leq \min\{2(\gamma - \eta_A), \eta_A\}$, under the equal sharing rule, in order to induce firm B to cooperate, firm A can make a transfer ω^* to firm B

conditional on the success of the R&D. Or equivalently, the equal sharing rule should be replaced by the unequal sharing rule v^* .

Under equal sharing rule, firm B will deviate. In order to achieve the beneficial mutual cooperation, unequal share or asymmetric distribution of the R&D benefit is necessary. And the less capable firm B actually obtains more. This is again the anti-Matthew effect.

4.2. Relational Contract

In the previous discussion, we assume that the R&D outcome is verifiable; thus, the formal contract is legally enforceable. However, the R&D outcome can be nonverifiable in many situations. For example, if the R&D outcome is some new innovation with intangible property, it may be quite difficult for a third party to verify it. Under such a circumstance, the formal contract is simply not available. In order to provide incentive for firm B to cooperate, the relational contract is an option (Levin, 2003).

We now consider the following cooperative strategy with transfer ω^* : The game starts in a cooperation phase in which both firms choose C . If the R&D is successful, firm A makes transfer ω^* to firm B. They continue doing so as long as 'good' outcome is observed and ω^* is made in the previous period. However, once 'bad' outcome is observed, firm A makes no transfer at that period. And from next period on, the game enters into a punishment phase in which firms play the stage-game Nash equilibrium (C, D) and firm A stops making transfer for T periods. After the punishment phase, a new cooperation phase starts again with both firms choosing C . In the case that the R&D is successful but firm A refuses to make transfer ω^* , the two firms play the stage-game Nash equilibrium (C, D) forever.

Although firms' choices for the RJV are unobservable, the transfer made or not is perfectly observable. As long as transfer is made, the punishment as specified in the cooperative strategy, playing (C, D) forever, will not be triggered. Along the equilibrium path, punishment will be triggered by failure in the RJV when firms follow the cooperative strategy.

Note that both firms have incentive to follow the strategy in the punishment phase because (C, D) is the stage-game Nash equilibrium. In the cooperation phase, the expected payoff vector $V_i^\omega = \{V_A^\omega, V_B^\omega\}$ can be written respectively as

$$V_A^\omega = (1 - \delta)[\gamma(\pi - \omega^*) - c] + \delta\gamma V_A^\omega + \delta(1 - \gamma)[(1 - \delta^T)(\eta_A\pi - c) + \delta^T V_A^\omega],$$

and

$$V_B^\omega = (1 - \delta)[\gamma(\pi + \omega^*) - c] + \delta\gamma V_B^\omega + \delta(1 - \gamma)[(1 - \delta^T)\eta_A\pi + \delta^T V_B^\omega].$$

Firm A can deviate from the strategy by stopping the transfer. The expected payoff for firm A to stop the transfer is

$$V_A^{\omega'} = (1 - \delta)(\gamma\pi - c) + \delta(\eta_A\pi - c).$$

Firm A can also deviate from the strategy by choosing D , and then the expected payoff for firm A is

$$V_A^{\omega''} = (1 - \delta)[\eta_B(\pi - \omega^*)] + \delta\eta_B V_A^\omega + \delta(1 - \eta_B)[(1 - \delta^T)(\eta_A\pi - c) + \delta^T V_A^\omega].$$

If firm B deviates from the strategy, the payoff is

$$V_B^{\omega'} = (1 - \delta)\eta_A(\pi + \omega^*) + \delta\eta_A V_B^\omega + \delta(1 - \eta_A)[(1 - \delta^T)\eta_A\pi + \delta^T V_B^\omega].$$

For cooperative strategy with transfer ω to be the equilibrium, the following conditions need to be satisfied: $V_A^\omega \geq V_A^{\omega'}$, $V_A^\omega \geq V_A^{\omega'}$, $V_B^\omega \geq V_B^{\omega'}$. From the above three incentive constraints, we can obtain respectively:

$$(1 - \delta)\gamma\omega^* \leq \delta\gamma + \delta^{T+1}(1 - \gamma)[V_A^\omega - (\eta_A\pi - c)], \quad (5)$$

$$(1 - \delta)[c - (\gamma - \eta_B)(\pi - \omega^*)] \leq \delta(\gamma - \eta_A) \quad (6)$$

$$(1 - \delta^T)[V_A^\omega - (\eta_A\pi - c)],$$

and

$$(1 - \delta)[c - (\gamma - \eta_A)(\pi + \omega^*)] \leq \delta(\gamma - \eta_A) \quad (7)$$

$$(1 - \delta^T)(V_B^\omega - \eta_A\pi).$$

The right-hand sides of Equations (5), (6), and (7) are strictly positive, whereas the left-hand sides of them go to 0 as $\delta \rightarrow 1$. Hence, fixing T , there exists $\hat{\delta} < 1$ such that one of Equations (5), (6), and (7) is just binding, whereas the other two are satisfied. Therefore, for all $\delta \geq \hat{\delta}$, the cooperative strategy constitutes a PPE of the repeated game.

Proposition 3:

There exists a pair $(\hat{\delta}, T)$ such that for all $\delta \geq \hat{\delta}$, cooperative strategy with transfer ω^* constitutes a PPE of the repeated game.

Note that in the unilateral deviation case, the more capable firm cooperates, whereas the less capable does not. When the R&D outcome is verifiable, the formal incentive contract can be used to motivate the less capable firm to cooperate. However, if the R&D outcome is nonverifiable, the self-enforcing relational contract is an alternative to induce the less capable firm to cooperate. When the two firms engage in a long-term relationship, they should be patient enough. In Proposition 3, we find the threshold discount factor $\hat{\delta}$ such that when δ is larger than or equal to $\hat{\delta}$, both firms follow the cooperative strategy specified above.

5. ENDOGENIZING THE SHARING RULE

The analysis in the previous sections is based on the equal sharing rule. This begs the question whether our result depends on this specific assumption or not. In this section, we endogenize the sharing rule and we assume that once the joint R&D is successful, firm A receives share s and firm B receives share $1-s$ of the total value 2π . The rest of the model remains the same as before.

Because of the new sharing rule, we rewrite the normal-form game in Section 2.1 as follows:

		Firm B	
		C	D
Firm A	C	$\gamma s 2\pi - c, \gamma(1-s)2\pi - c$	$\eta_A s 2\pi - c, \eta_A(1-s)2\pi$
	D	$\eta_B s 2\pi, \eta_B(1-s)2\pi - c$	0, 0

It is easy to show that in the one-shot game, given $\frac{c}{\pi} \leq \frac{2\eta_A(\gamma-\eta_A)}{\gamma}$, when $s \in \left[\frac{c}{2\pi\eta_A}, 1 - \frac{c}{2\pi(\gamma-\eta_A)}\right]$, (C, C) is the Nash equilibrium. However, when $\frac{2\eta_A(\gamma-\eta_A)}{\gamma} < \frac{c}{\pi} < \gamma$, we cannot find any sharing rule to support (C, C) as an equilibrium outcome in the one-shot game.

As before, the two firms can engage in repeated interactions to support the mutual cooperation. For simplicity, we focus on the grim trigger strategy: Given the sharing rule $S \equiv (s, 1-s)$, both firms start the game by choosing C for the RJV. They keep doing so as long as 'good' outcome is observed in the previous period. Once 'bad' outcome is observed, both firm choose D forever.

We can write the equilibrium payoffs V_A^S and V_B^S as

$$V_A^S = (1-\delta)(\gamma s 2\pi - c) + \delta \gamma V_A^S,$$

and

$$V_B^S = (1-\delta)[\gamma(1-s)2\pi - c] + \delta \gamma V_B^S.$$

If firm A deviates, its payoff is

$$V_A^{S'} = (1-\delta)\eta_B s 2\pi + \delta \eta_B V_A^S.$$

If firm B deviates, its payoff is

$$V_B^{S'} = (1-\delta)\eta_A(1-s)2\pi + \delta \eta_A V_B^S.$$

For the grim trigger strategy to be an equilibrium, we need $V_A^S \geq V_A^{S'}$ and $V_B^S \geq V_B^{S'}$. From $V_A^S \geq V_A^{S'}$, we obtain

$$\delta \geq \frac{c - (r - \eta_B)s 2\pi}{c\eta_B}. \quad (8)$$

From $V_B^S \geq V_B^{S'}$, we obtain

$$\delta \geq \frac{c - (r - \eta_A)(1-s)2\pi}{c\eta_A}. \quad (9)$$

Combining Equations (8) and (9), we find that given $\delta \geq \delta'$, where δ' is defined for Proposition 2, there exists $s^* \in \left[\frac{c(1-\delta\eta_B)}{2\pi(\gamma-\eta_B)}, 1 - \frac{c(1-\delta\eta_A)}{2\pi(\gamma-\eta_A)}\right]$ such that the above grim trigger strategy with sharing rule $S^* \equiv (s^*, 1-s^*)$ can be supported as a PPE. In addition, as long as $\delta' \leq \delta < \delta_B^*$, we will have $(1-s^*) - s^* > 0$, which means the less capable firm receives a larger share than the more capable one. Therefore, we can obtain the following Proposition:

Proposition 4:

Given $\delta \geq \delta'$, there exists $s^* \in \left[\frac{c(1-\delta\eta_B)}{2\pi(\gamma-\eta_B)}, 1 - \frac{c(1-\delta\eta_A)}{2\pi(\gamma-\eta_A)}\right]$ such that the grim trigger strategy with sharing rule S^* can be supported as a PPE. Furthermore, when $\delta' \leq \delta < \delta_B^*$, the less capable firm receives a larger share, or $(1-s^*) - s^* > 0$.

Proposition 4 finds the endogenous sharing rule S^* that can support the mutual cooperation. It confirms our previous result that given $\delta' \leq \delta < \delta_B^*$, the less capable firm receives a larger share, and it also indicates that this result does not depend on the assumption of the equal sharing rule.

6. CONCLUSION

It is difficult to maintain cooperation among asymmetric firms because some are more likely to deviate than others. Within a repeated game theoretical framework with imperfect monitoring, this article studies how to support R&D cooperation between firms with different technological capabilities. We show that long-term relationship combining with transfer can help to maintain the mutual cooperation when the discount factor is not large enough. Furthermore, previous papers point out the Matthew effect, but this article obtains the opposite effect, the anti-Matthew effect. We find that under certain condition, the more capable firm can use transfer to bribe the less capable to cooperate, and both firms are better off. But the less capable firm receives more. Although our model deals with the asymmetry in technological capabilities, it can be extended to incorporate other asymmetries such as sizes, costs, and so on. The underlying implications of our results can also be generalized to any cooperation between asymmetric partners.

There are limitations of this research. For example, for simplicity, we only consider the trigger strategies. But there are many other more sophisticated strategy profiles that can support the cooperative equilibria. The two-action, two-outcome environment of our model is also quite simple. It is ready to be extended to more general settings.

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NOTES

1. Merton (1968) first coins the term 'the Matthew effect'. And its name comes from a line in the biblical Gospel of Matthew: For to everyone who has, more shall be given, and he will have an abundance; but from the one who does not have, even what he does have shall be taken away.—Matthew 25:29, New American Standard Bible.
2. Veugelers and Kesteloot (1996) use the term 'the Matthew effect' in their paper's title.
3. Other papers, such as that of Sinha and Cusumano (1991), also make the same assumption.
4. For example, the R&D cooperation between Microsoft and Nokia launched recently is highly complementary.
5. One possible interpretation for the equal sharing rule is that the two firms have equal positions in the product

market. When they both adopt the newly invented technology for their productions, they obtain equal benefits from the R&D result.

6. This technical assumption is to set the cutoffs for the following three cases. We can assume $\gamma \geq 2\eta_A$ instead, and then we only need to switch the cutoffs in those three cases.
7. In reality, many RJVs have indeterminate end dates once set up, which fits the infinite horizon repeated game framework. SEMATECH, for example, launched in 1987, has succeeded in many R&D projects and still engaged many original partners such as Intel, AMD, HP, IBM, and Texas Instruments.
8. For detailed discussion of perfect public equilibrium, see Fudenberg, Levine and Maskin (1994).
9. Duso, Pennings and Seldeslachts (2010) find that firms frequently enter and exit the research consortia after their initial formation.
10. The reason that we consider the grim trigger strategy with transfer is the following. δ_B^* is the lowest possible discount factor to support the cooperative equilibrium, which we find by using the grim trigger strategy without transfer. Given $\delta < \delta_B^*$, it is proper to keep the grim trigger strategy in order to find the lowest possible transfer to support the mutual cooperation outcome. If not, one can imagine the transfer will be much higher so that the mutual cooperation is harder to maintain.

APPENDIX A

Proof of Proposition 1

First, note that both firms have incentive to stick to the strategy in the punishment phase because it is a best response to play D . In the cooperation phase, the expected payoff for firm i is

$$V_i = (1 - \delta)(\gamma\pi - c) + \delta\gamma V_i + \delta^{T+1}(1 - \gamma)V_i.$$

However, if firm i deviates from the strategy by playing D , whereas firm j follows the equilibrium strategy, firm i 's expected payoff would be:

$$V'_i = (1 - \delta)\eta_j\pi + \delta\eta_j V_i + \delta^{T+1}(1 - \eta_j)V_i.$$

Hence, the incentive constraint for firm i not to deviate is

$$V_i \geq V'_i.$$

Or, the incentive constraints for firm A and firm B are

$$(1 - \delta)[c - (\gamma - \eta_B)\pi] \leq \delta(\gamma - \eta_B)(1 - \delta^T)V_A, (IC_A)$$

and

$$(1 - \delta)[c - (\gamma - \eta_A)\pi] \leq \delta(\gamma - \eta_A)(1 - \delta^T)V_B, (IC_B)$$

respectively.

Because $\eta_A > \eta_B$ and $V_A = V_B$, (IC_B) implies (IC_A) . Therefore, we only need to consider (IC_B) . The right-hand side of (IC_B) is strictly positive, whereas the left-hand side goes to 0 as $\delta \rightarrow 1$. Hence, fixing T , there exists $\bar{\delta} < 1$ such that (IC_B) is binding. And for all $\delta \geq \bar{\delta}$, the cooperative strategy constitutes a PPE of the repeated game. $\square$

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